

Name: _____

Date: _____

STATISTICS, (MEAN, MEDIAN, MODE AND RANGE)

LO: I can calculate the mean, median, mode and range from a set of data.

Averages and range

The **mean** is the sum of values divided by the count. The **median** is the middle value when data is ordered. The **mode** is the most frequent value.

Range = highest value minus lowest value. It measures the spread of the data.

Quick examples

●	Mean of 2, 4, 6 = $12 \div 3 = 4$	sum $\div$ count
●	Median of 3, 5, 7, 9, 11 = 7	middle value
●	Mode of 2, 3, 3, 5, 7 = 3	most frequent
●	Median of 2, 4, 6, 8 = 6	should be $(4+6) \div 2 = 5$
●	Range of 3, 7, 10, 2 = 10	range = $10 - 2 = 8$

Key idea

Mean = total $\div$ number of values. Median = middle value (or mean of two middle values for even counts).

$$\text{Mean} = x \div n$$

Sum all values and divide by how many there are

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - finding the mean

Find the mean of 5, 8, 3, 12, 7.

$$\text{Sum} = 5 + 8 + 3 + 12 + 7 = 35$$

$$\text{Count} = 5$$

$$\text{Mean} = 35 \div 5 = 7$$

$$\text{Mean} = 7$$

Add all values then divide by how many there are.

Example 2 - finding the median

Find the median of 9, 3, 7, 1, 5, 8, 2.

Order: 1, 2, 3, 5, 7, 8, 9

7 values, middle is the 4th

$$\text{Median} = 5$$

$$\text{Median} = 5$$

Always put the data in order first.

Example 3 - even number of values

Find the median of 4, 7, 10, 13.

Data is already ordered

Two middle values: 7 and 10

$$\text{Median} = (7 + 10) \div 2 = 8.5$$

$$\text{Median} = 8.5$$

For an even count, average the two middle values.

Example 4 - from a frequency table

Scores: 1(freq 2), 2(freq 5), 3(freq 8), 4(freq 3), 5(freq 2). Find the mean.

$$\text{Total} = 1 \times 2 + 2 \times 5 + 3 \times 8 + 4 \times 3 + 5 \times 2$$

$$= 2 + 10 + 24 + 12 + 10 = 58$$

$$\text{Total freq} = 2 + 5 + 8 + 3 + 2 = 20$$

$$\text{Mean} = 58 \div 20 = 2.9$$

$$\text{Mean} = 2.9$$

Multiply each value by its frequency, then divide by total frequency.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding the mean

Calculate the mean of each data set.

- a) 4, 6, 8, 10, 12
- b) 3, 7, 5, 9, 11
- c) 15, 20, 25, 30
- d) 1, 1, 2, 3, 5, 8

2. Median and mode

Find the median and mode.

- a) 3, 5, 5, 7, 9
- b) 2, 4, 6, 8, 10, 12
- c) 1, 3, 3, 5, 7, 7, 9
- d) 10, 20, 20, 30, 40

3. Range and comparing data

Find the range. Compare the data sets.

- a) 5, 8, 12, 3, 15
- b) 22, 25, 21, 28, 24
- c) Set A: mean 50, range 20. Set B: mean 50, range 5. Which is more consistent?
- d) 100, 102, 98, 105, 95

4. From frequency tables

Calculate the mean from each frequency table.

- a) Score 1($f=3$), 2($f=5$), 3($f=7$), 4($f=4$), 5($f=1$)
- b) Goals 0($f=4$), 1($f=6$), 2($f=3$), 3($f=2$)
- c) Shoe size 5($f=2$), 6($f=5$), 7($f=8$), 8($f=3$), 9($f=2$)
- d) Hours 0($f=1$), 1($f=4$), 2($f=6$), 3($f=5$), 4($f=4$)

5. Mixed averages

Solve each problem.

- a) Mean of 5 numbers is 8. Sum?
- b) 6 numbers have mean 10. A 7th number (17) is added. New mean?
- c) Data: 3, 5, x , 9, 11. Mean is 7. Find x .
- d) Median of 2, 5, x , 12, 15 is 8. Find x .
- e) Range is 12, lowest value is 5. Highest value?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Mean = sum of values ÷ number of values ☐

Correct answer: _____

Reason: _____

b) The median of 2, 4, 6, 8 is 6 ☐

Correct answer: _____

Reason: _____

c) The mode can be more than one value ☐

Correct answer: _____

Reason: _____

d) Range = highest value ☐

Correct answer: _____

Reason: _____

e) For an even set, the median is the mean of the two middle values ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A class of **28 students** has a mean test score of **65**. Two students who scored **90 and 85** leave. What is the new mean score for the remaining students? Round to 1 decimal place.

Expression: _____

Simplified: _____

b) Five numbers have a mean of **12**, a median of **11**, a mode of **9**, and a range of **10**. Find all five numbers.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

